



DAAN862: ANALYTICS PROGRAMMING IN PYTHON

Lesson 5: Data Wrangling with Pandas

Lesson 5.3: Reshaping and Pivoting

There are a number of operations to rearrange tabular data. These are referred to as reshaping or pivoting.

(click titles to learn more)

Reshaping with Hierarchical Indexing

Hierarchical Indexing provides a convenient way to rearrange data.

There are two primary actions:

stack: This "rotates" or pivots from the columns in the data to the rows

unstack: This pivots from the rows into the columns



Examples

Let's use a pivot-table like DataFrame with row and column indexes as shown in Figure 5.28:

Click the dropdown to view Figure 5.28:

```
In [50]: data = pd.DataFrame(np.arange(6).reshape((2, 3)),
...:                        index=pd.Index(['Ohio', 'Colorado'], name='state'),
...:                        columns=pd.Index(['one', 'two', 'three'],
...:                                       name='number'))
...: data
Out[50]:
```

number	one	two	three
state			
Ohio	0	1	2
Colorado	3	4	5

Fig 5.28 (click to enlarge)

Applying the stack method to this data converts the tree columns into an inner-level index and returns a Series as shown in Figure 5.29:

Click the dropdown to view Figure 5.29

```
In [51]: result = data.stack()

In [52]: result
Out[52]:
```

state	number	
Ohio	one	0
	two	1
	three	2
Colorado	one	3
	two	4
	three	5

```
dtype: int32

In [53]: type(result)
Out[53]: pandas.core.series.Series
```

```
In [54]: result.index
Out[54]:
MultiIndex(levels=[['Ohio', 'Colorado'], ['one', 'two', 'three']],
            labels=[[0, 0, 0, 1, 1, 1], [0, 1, 2, 0, 1, 2]],
            names=['state', 'number'])
```

Fig 5.29 (click to enlarge)